

THE SECRETARY OF DEFENSE
WASHINGTON

8 May 1961

Dear Mr. Vice President:

Attached to this letter is a report entitled "Recommendations for Our National Space Program: Changes, Policies, Goals", dated 8 May 1961. This document represents our joint thinking. We recommend that, if you concur with its contents and recommendations, it be transmitted to the President for his information and as a basis for early adoption and implementation of the revised and expanded objectives which it contains.

Very respectfully,

/s/ James E. Webb

James E. Webb
Administrator
National Aeronautics and
Space Administration

/s/ Robert S. McNamara

Robert S. McNamara
Secretary of Defense

1 Inclosure
Report

The Vice President

United States Senate

This document is downgraded to
UNCLASSIFIED
upon removal of inclosure

Control #
61-2472

REPORT

RECOMMENDATIONS FOR OUR NATIONAL SPACE PROGRAM:

CHANGES, POLICIES, GOALS

TABLE OF CONTENTS	Page
I - Recommendations for FY 1962 Add-Ons	1
II - National Space Policy	7
III - Major National Space Goals	13
Tab A - The Soviet Program and Capabilities	17
Tab B - U. S. Space Program	19
Tab C - Tables:	
Table 1 - Space Activities of the United States Government - New Obligation Authority/ Program Basis - in millions	27
Table 2 - National Aeronautics and Space Administration FY 1962 Budget and Proposed Add-Ons	28
Table 3 - Department of Defense Space and Related Programs FY 1962 Budget and Proposed Add-Ons	29
Table 4 - Estimated Projections of N.A.S.A. and Defense Space Programs (Excludes A.E.C., Weather Bureau, and N.S.F.)	30

EXCLUDED FROM AUTOMATIC REGRADING;
DOD DIR 5200.10 DOES NOT APPLY

INTRODUCTION

It is the purpose of this report (1) to describe changes to our national space efforts requiring additional appropriations for FY 1962; (2) to outline the thinking of the Secretary of Defense and the Administrator of NASA concerning U. S. status, prospects, and policies for space; and (3) to depict the chief goals which in our opinion should become part of an Integrated National Space Plan. These matters are covered in Sections I, II, and III, respectively.

Three appendices (Tabs A through C) support these sections. Tab A highlights the Soviet space program. The bulk of this Tab (Attachment A) is separated from this report since it bears a special security classification. Tab B includes a description of major U. S. space projects and elements. Tab C provides financial summaries of the present programs, the proposed add-ons, and the future costs of the programs.

This first joint report contains the results of extensive studies and reappraisals. It is a first and not our last report and does not, of course, represent a complete or final word about our space undertakings.

SECRET

I. RECOMMENDATIONS FOR FY 1962 ADD-ONS

Our recommendations for additional FY 1962 NOA for our space efforts are listed below. They total \$626 million of which all but \$77 million is for NASA. Certain of these additions will accelerate projects which need to be accomplished more quickly if national space goals are to be reached on time. Other additions augment projects or programs to afford greater likelihood of success or to acquire concurrently data needed to implement long range goals for which we would otherwise have had to wait. Some of the additions, especially those for rocket engine, booster, and upper stage developments, support parallel programs to insure that the failure or delay in a single launch vehicle development will not place our long range goals in jeopardy. It is our belief that it is feasible to accomplish these objectives of acceleration, augmentation and greater certainty of success through the application of the funds specified. The general objectives and their implementation in each particular case are amply supported by the investigations and assessments of qualified scientific and technical people intimately familiar with our space undertakings in detail and in the large.

These FY 1962 additions represent a vital first stage funding toward implementing longer range goals, including the objective of manned lunar exploration in the latter part of this decade, specified in Section III. They will, we believe, prove to be consistent with the objectives and policies set forth in Section II.

a. Spacecraft for Manned Lunar Landing and Return

To achieve the goal of landing a man on the moon and returning him to earth in the latter part of the current decade requires immediate initiation of an accelerated program of spacecraft development.

The program designated Project Apollo includes initial flights of a multi-manned orbiting laboratory to qualify the

spacecraft, and manned flights around the moon before attempting the difficult lunar landing.

The additional funds requested will be used to extend tests with the Mercury spacecraft to learn more about the behavior of man during longer duration flights in space, to study the biomedical problems encountered in outer space, to investigate the problems of reentering the earth's atmosphere from lunar return speeds, to initiate the development of the multi-manned spacecraft for the mission, and for ground tracking and other facilities.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'L</u>
FUNDS:	29.5M	240.0M	210.5M

b. Launch Vehicle Development

1. For the manned lunar landing

The advanced goal of manned landing on the moon requires the development of a launch vehicle (Nova) with a thrust of about six times greater than that of the largest vehicle now under development (Saturn). The funds requested are to accelerate the development of 1½ million pounds thrust liquid fueled rocket engine now under development; for design, engineering, and component development of the Nova vehicle; and to initiate the construction of necessary facilities required in support of the vehicle development and test.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'L</u>
FUNDS:	42.7M	155.2M	112.5M

2. Solid propellant parallel approach

To assure a high degree of success in achieving the manned lunar landing goal a parallel approach to the development of a first stage for the very large launch vehicle (NOVA) must be undertaken. In addition to the use of liquid fuels for this purpose we must also undertake the immediate development of large solid rocket launch vehicles. When developments in both liquid fueled and solid fueled rockets have

SECRET

progressed to a stage where one or the other can be shown to be the superior approach, it will be pursued as the principal launch vehicle development. Certain elements of the solid rocket development are also believed to have future military importance. The DoD will be responsible for this development, being fully responsive to NASA requirements.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS(DOD):	-0-	62M	62M

c. Development of An Upper Stage for Titan II

The Atlas-Agena combination is the most powerful launch vehicle available to the U. S. until the Atlas-Centaur becomes operational in 1963. The Atlas-Agena cannot place more than 5,000 pounds in a 300-mile orbit. The Atlas-Centaur, if successful, will be capable of launching as much as 8,500 pounds.

Because of an urgent need by both DoD and NASA of a vehicle with the performance of Atlas-Centaur, a back-up for this launch vehicle is considered essential. Therefore it is proposed to initiate development of an upper stage for Titan II which will then provide a strong back-up for the Atlas-Centaur. The Titan II upper stage development will be terminated if the timely success of Atlas-Centaur becomes apparent.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS(DOD):	-0-	15M	15M

d. Unmanned Lunar Exploration

Before attempting a manned lunar landing, it is essential to learn more about the phenomena that exist in space near the moon and about the nature of the characteristics of the moon's surface. The programs now under way, designated Ranger and Surveyor, are designed to provide this information. With the additional funds, the number of flights in these programs will be increased and the program will be accelerated to provide timely information and greater assurance.

~~SECRET~~

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS:	71.67M	134.67M	63.0M

e. Scientific Experiments on Space Environment

Knowledge of the space environment through which man must travel to the moon is now still very meager. The nature and characteristics of the radiation emanating from the sun and from outer space must be thoroughly studied and understood. Man's ability to survive in the outer space and on the surface of the moon depends on this knowledge. The additional funds will augment and expedite current programs that will provide this information.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS:	72.2M	87.2M	15.0M

f. Satellite Communications System

With the launching of the Echo and Courier communications satellites, the United States achieved a position of leadership in the use of satellites for worldwide communications. Studies by many qualified organizations have shown the potential economic, political, and military value of these systems. Programs are currently underway to exploit these advantages. The funding requested will accelerate the developments, lead to a much earlier availability of an operational system, and maintain the United States position of leadership.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS:	44.6M	94.6M	50.0M

g. Meteorological Satellites for Worldwide Weather Prediction*

The outstanding success demonstrated by the U. S. Tiros weather satellite in worldwide weather prediction will enable the U. S. to exploit this technology for the benefit of all mankind.

* The Department of Commerce is considering the request of an additional 53.5M in FY 62 to initiate an operational meteorological satellite system.

SECRET

Developments are now under way which will lead to operational systems. To assure success of an operational system at the earliest time it is necessary to augment the funding of this program by the amount shown below.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS:	28.2M	50.2M	22.0M

h. Nuclear Rocket Development*

Nuclear rocket development (Rover) must be carried out on an accelerated basis because of its great potential for even more difficult missions than landing a man on the moon. This program faces many difficult technological problems which will require substantial support over a number of years for their solution. The funding will provide for augmented research and development and essential facilities for the conduct of the program.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS:	17.5M	40.5M	23.0M

i. Supporting Research and Technology

The successful accomplishment of the accelerated program goals requires immediate expansion of basic researches and advances in technology in many fields. For the conduct and management of this major effort some additional increase in the NASA staff will also be required. Requirements for the additional research, advancement of technology and support of personnel and plant are shown below.

	<u>APPROVED BUDGET</u>	<u>RECOMM. BUDGET</u>	<u>ADD'AL</u>
FUNDS:	393.5M	446.5M	53.0M

* The Atomic Energy Commission has requested an additional 7M in FY 62 to support the reactor portion of the Nuclear Rocket Development Program.

~~SECRET~~

II. NATIONAL SPACE POLICY

The recommendations made in the preceding Section imply the existence of national space goals and objectives toward which these and other projects are aimed. Major goals are summarized in Section III. Such goals must be formulated in the context of a national policy with respect to undertakings in space. It is the purpose of this Section to highlight our thinking concerning the direction that such national policy needs to take and to present a backdrop against which more specific goals, objectives and detailed policies should, in our opinion, be formulated.

a. Categories of Space Projects

Projects in space may be undertaken for any one of four principal reasons. They may be aimed at gaining scientific knowledge. Some, in the future, will be of commercial or chiefly civilian value. Several current programs are of potential military value.

Finally, some space projects may be undertaken chiefly for reasons of national prestige.

The U. S. is not behind in the first three categories. Scientifically and militarily we are ahead. We consider our potential in the commercial/civilian area to be superior. The Soviets lead in space spectacles which bestow great prestige. They lead in launch vehicles needed for such missions. These bestow a lead in capabilities which may some day become important from a military point of view. For these reasons it is important that we take steps to insure that the current and future disparity between U. S. and Soviet launch capabilities be removed in an orderly but timely way. Many other factors however, are of equal importance.

b. Space Projects for Prestige

All large scale space projects require the mobilization of resources on a national scale. They require the development and successful application of the most advanced technologies. They call for skillful management, centralized control and unflagging pursuit of long-range

goals. Dramatic achievements in space, therefore, symbolize the technological power and organizing capacity of a nation.

It is for reasons such as these that major achievements in space contribute to national prestige. Major successes, such as orbiting a man as the Soviets have just done, lend national prestige even though the scientific, commercial or military value of the undertaking may by ordinary standards be marginal or economically unjustified.

This nation needs to make a positive decision to pursue space projects aimed at enhancing national prestige. Our attainments are a major element in the international competition between the Soviet system and our own. The non-military, non-commercial, non-scientific but "civilian" projects such as lunar and planetary exploration are, in this sense, part of the battle along the fluid front of the cold war. Such undertakings may affect our military strength only indirectly if at all, but they have an increasing effect upon our national posture.

c. Planning

It is vital to establish specific missions aimed mainly at national prestige. Such planning must be aimed at both the near-term and at the long range future. Near-term objective alone will not suffice. The management mechanisms established to implement long range plans must be capable of sustained centralized direction and control. An immediate task is to specify long-range goals, to describe the missions to be accomplished, to define improved management mechanisms, to select the launch vehicles, the spacecraft, and the essential building blocks needed to meet mission goals. The long-term task is to manage national resources from the national level to make sure our goals are met.

It is absolutely vital that national planning be sufficiently detailed to define the building blocks in an orderly and integrated way. It is absolutely vital that national management be equal to the task of focusing resources, particularly scientific and engineering manpower

~~SECRET~~

resources, on the essential building blocks. It is particularly vital that we do not continue to make the error of spreading ourselves too thin and expect to solve our problems through the mere appropriation and expenditure of additional funds.

d. Feasibility

It is technically feasible to match and surpass the Soviets in all areas of national space competition whether scientific, commercial, military or in the area of national prestige. Certain steps need to be undertaken right away. Those requiring additional appropriations in FY 1962 were described in Section I.

Additional important actions have also been defined. They include the necessity to specify standardized "work horse" building block combinations to support our space efforts for the long pull. It is particularly important to define building block combinations of boosters and upper stages for each major class of payload and mission. Conversely, it is important to avoid wherever possible the creation of complex and costly launch vehicles and other equipments optimized for and largely limited in application to a single project or mission. This principle has been recognized and applied to portions of our launch-vehicle developments. It has not been applied to all. It must govern all our efforts in the future. Major sub-elements including guidance systems, control systems, power supplies, telemetry, recovery and other basic system elements must also be standardized and used repetitively to the maximum possible degree.

After fully adequate study we must specify the minimum family of launch vehicle systems that will enable us to accomplish both near term (such as communication satellites) and long range missions (such as lunar or planetary exploration) which will comprise our national space goals.

If properly conceived and designed the vehicles will be used for many years in dozens and perhaps hundreds of costly missions. Their development and procurement will involve the expenditure of billions of

dollars over a period of a decade or more. It is essential that we set out foot on the correct path leading to the future. The decisions made in the beginning will define that path. Once embarked upon it, we cannot turn back or turn aside without losing time which can never be regained.

e. Background information bearing on the problem:

These words would not be written and this report would not be called for if we were satisfied with our status and our prospects for the future in space. We are not satisfied. We are behind in important ways and it is not clear that we are catching up. In reading the balance of this report, it is important to have in mind some of the highlights of space history, of our posture today, and of our prospects for the future. It is important to realize that more money is not the only answer. While considerable additional effort will be called for, a principal problem is how better to harness, not merely how further to expand, the human and physical resources already at hand.

It is important to note that the recent Soviet attainments are the result of a program planned and executed at the national level over a long period of time. The decisions which led to the current successes were, for the most part, made many years ago. Many of them, in fact, must have been made in the early part of the 1950-1960 decade.

That decade has witnessed a great expansion in U.S. government-sponsored research and development, especially for large-scale defense programs. Enormous strides have been made, particularly in our space efforts and in the development of related ballistic missile technology on a "crash" basis. We have, however, incurred certain liabilities in the process. We have ^{over}encouraged the development of entrepreneurs and the proliferation of new enterprises. As a result, key personnel have been thinly spread. The turnover rate in U.S. defense and space industry has had the effect of removing many key scientific engineering personnel from their jobs before the completion of the projects for which they were employed. Strong concentrations of technical talent needed for the best

work on difficult tasks have been seriously weakened. Engineering costs have doubled in the past ten years.

These and other trends have a strong adverse effect on our capacity to do a good job in space. The inflation of costs has an obvious impact and they are still rising at the rate of about seven per cent per year. This fact alone affects forward planning. It has often led to project stretch-outs, and may again in future years. The spreading out of technological personnel among a great many organizations has greatly slowed down the evolution of design and development skills at the working level throughout the country. Precisely the opposite is true in the USSR, where the turnover rate is very low and where skilled cadres of development personnel remain in existence for a great many years.

It is not suggested that we apply Soviet type restrictions and controls upon the exercise of personal liberty and freedom of choice. It is suggested, however, that our American system can be and must be better utilized in the future than in the past.

Our space efforts, like many of our military weapons developments, have suffered because of our tendency to "improve" and to embellish our designs. We have allowed ourselves to strive for the optimum solution to nearly every problem project-by-project. We have often tried to "integrate" very complex system elements at minimum weight and with very little margin for safety or for error. Many have come to think that such techniques are the natural and obvious way to get jobs done. They are not, they will not succeed and they must be changed.

We must address ourselves to these problems more effectively in the future than in the past. We must create mechanisms to lay out and to insist upon achievement, not mere improvement. We must stress performance, not embellishment. We must insist from the top down, that, as the Russians say, "the better is the enemy of the good".

~~SECRET~~

f. Summary.

Clearly, then, the future of our efforts in space is going to depend on much more than this year's appropriations or tomorrow's new idea. It is going to depend in large measure upon the extent to which this country is able to establish and to direct an "Integrated National Space Program".

True, it will be necessary to support our space efforts at a higher funding level than recommended before. Such support will have to be backed by the Administration, by the Congress, and by the American people. If, however, the application of more money leads to still further cost increases and still further thinning out of technical manpower and technical supervision, it is likely that the Russians will be ahead of us ten years from now just as they are today.

It will be necessary, therefore, to find a way to formulate and apply plans and policies aimed at insuring the success of an Integrated National Space Program. Top level scientific and policy direction must be forthcoming from the top management echelon. The mere statement of broad objectives will not be good enough. Periodic budget reviews and their intensification in the spring of each year will not suffice. It will be necessary to impose policy and management actions which will alter many of the trends of the past ten years, particularly in the management of research and engineering resources on a national scale. It will be necessary to impose actions which may involve painful cancellations and redirections.

These and other policies, too, must be supported by the Administration, by the Congress, and by the American people to insure success for the long pull ahead.

Our joint efforts are addressed to the creation of management tools to deal with these and other problems we will face in the years ahead.

III. MAJOR NATIONAL SPACE GOALS

It is the purpose of this section to outline some of the principal goals, both long range and short range, toward which our national space efforts should, in our opinion, be directed. It is not the intent to specify all of the goals or even all of the major goals of importance to a National Space Plan. We wish to stress five principal objectives which in our opinion have not been adequately formulated or accepted in the past and which we believe should be accepted as a basis for specific project undertakings in the years ahead.

a. Manned Lunar Exploration

We recommend that our National Space Plan include the objective of manned lunar exploration before the end of this decade. It is our belief that manned exploration to the vicinity of and on the surface of the moon represents a major area in which international competition for achievement in space will be conducted. The orbiting of machines is not the same as the orbiting or landing of man. It is man, not merely machines, in space that captures the imagination of the world.

The establishment of this major objective has many implications. It will cost a great deal of money. It will require large efforts for a long time. It requires parallel and supporting undertakings which are also costly and complex. Thus, for example, ^{the} RANGER and the SURVEYOR Projects and the technology associated with them must be undertaken and must succeed to provide the data, the techniques and the experience without which manned lunar exploration cannot be undertaken.

The Soviets have announced lunar landing as a major objective of their program. They may have begun to plan for such an effort years ago. They may have undertaken important first steps which we have not begun.

It may be argued, therefore, that we undertake such an objective with several strikes against us. We cannot avoid announcing not only our general goals but many of our specific plans, and our successes

~~SECRET~~

and our failures along the way. Our cards are and will be face up--their's are face down.

Despite these considerations we recommend proceeding toward this objective. We are uncertain of Soviet intentions, plans or status. Their plans, whatever they may be, are not more certain of success than ours. Just as we accelerated our ICBM program we have accelerated and are passing the Soviets in important areas in space technology. If we set our sights on this difficult objective we may surpass them here as well. Accepting the goal gives us a chance. Finally, even if the Soviets get there first, as they may, and as some think they will, it is better for us to get there second than not at all. In any event we will have mastered the technology. If we fail to accept this challenge it may be interpreted as a lack of national vigor and capacity to respond.

b. Worldwide Operational Satellite Communication Capability

It is our belief that advances in technology will make it possible to set up an operational satellite-based telecommunications capability within a few years. It is too early to be sure what kind of capability we should create.

We are certain, however, that at least one of several current possibilities is very likely to prove successful and practical. We are also confident that an operational communication satellite capability can have far reaching applications and implications for the U. S.

Many commercial enterprises in this country have displayed great interest in this subject. It is virtually certain that communication satellites will have commercial utility in future years. The Department of Defense is keenly interested in view of its large use of commercial capability and is also undertaking the development (Project ADVENT) of a satellite aimed principally at fulfilling military requirements.

Finally and perhaps most important of all, communication satellites uniquely provide a way of relaying information from one point

to another over great distances. A successful global satellite-based communication system may be looked upon somewhat as an overhead cable thousands of miles in the air to which users all over the world may make connections by means of invisible radio beams.

Accordingly, we have recommended that the NASA Communication Satellite effort be expanded in FY 1962. Within the next few years this country which is already the world's leader in communications of all kinds will be able to deploy a worldwide satellite-based communication capability.

c. Worldwide Operational Satellite Weather Prediction System.

The TIROS I meteorological satellite operated for several months, transmitting pictures of cloud cover around the world. TIROS II, launched last November, is still transmitting cloud pictures and infrared data. The Weather Bureau, the NASA, and other interested governmental agencies are closely coordinating their interests with respect to TIROS and follow-on meteorological satellites. A worldwide system of such satellites will be of great value to people in every country, to public and private interests in the U. S., and to our military forces, particularly those at sea and in the air. The addition of \$22 million in FY 1962 will accelerate the early attainment of preliminary operational capability.

d. Scientific Investigation

Fundamental to and underlying all progress in the exploration and application of space is the knowledge to be gained from the space sciences. It is essential that the national space sciences program be broad and comprehensive both in content and in participation by the scientific community of the world.

Before man can explore in the vast and hostile regions of space more knowledge is required on the effects of hard vacuum, weightlessness, and radiation.

The broad program that is recommended includes the following objectives:

SECRET

- To understand the nature of the sun and its influence on the earth.
- To investigate the solar system, its nature, and its history.
- To search for life in the solar system.
- To study cosmology, the history and nature of the universe.

Researchers in government laboratories, universities, industry, and other scientific organizations must participate in the space sciences effort and the space science program should support not only the existing talent but also the development of new talent. This requires support of universities in the education of the young scientists who are inspired by the challenge of space and who will strengthen the program with their vitality and new ideas.

For the fiscal year 1962 we recommend adding fifteen million dollars to the program to assure success of these objectives.

e. Large Scale Boosters for Potential Military Use.

Space technology is in its infancy. The first U. S. satellite was launched only 3-1/2 years ago. Vast resources are presently devoted to this field in the Communist world and in our own. The future potentialities and capabilities that space technology will afford cannot be foreseen. Their military potential and implications are largely unknown. It is certain, however, that without the capacity to place large payloads reliably into orbit, our nation will not be able to exploit whatever military potential unfolds in space.

We believe it is important, therefore, to insure that large scale boosters are made available. They should, of course, form part of a family of launch vehicles applicable to many missions and thoroughly integrated with NASA developments and with characteristics suitable for NASA needs.

We have agreed jointly in recommending the addition of \$62 million for the DOD in FY 1962 to undertake the development of large scale solid propellant rocket motors.

SECRET

PAB A: THE SOVIET PROGRAM AND CAPABILITIES

Attachment A depicts the Soviet space program in considerable detail. It is separated from this report for security reasons, but it should be read by the key policy makers concerned with the U. S. space program. It should be read not only because of what it tells about the Soviet program in space, but for what it reveals concerning the caliber of the competition we are up against in this important arena of the cold war.

The U. S. did not undertake a corresponding planning effort at the national level until much later. Space was not recognized as an area of importance to be planned for and pursued until after Sputnik I. Great efforts were then made to utilize the tools at hand to enter the space arena. Early projects were undertaken on a crash basis. New governmental and industrial organizations were formed and expanded under pressure and in haste. For several years it was necessary to "make do" with too-small boosters and launch vehicles employing devices developed and deployed with unusual urgency. However dedicated or effective such efforts were, they were not the product of a deliberate effort adhering to preplanned schedules and objectives.

SECRET

The U. S. space program in the past three years reflects this situation in many ways. We have been forced to design with inadequate margin for error or deficiencies in thrust. We have been forced to develop elaborate and often unreliable new ways to cram complex equipments in a very small space. Our results have, despite many excellent achievements, been disappointing in many important ways. Nearly half of our attempted launchings failed to achieve orbit. Certain programs achieved success, real success, on fewer than a third of all attempts. To a large degree, though not entirely, of course, these disappointments are symptoms of the lack of adequate national planning and guidance for the long pull.

It is prudent to suppose that the next decade will be marked with Soviet achievements in space which will be well planned, well directed, and executed with deliberateness and skill.

SECRET

TAB B: U. S. SPACE PROGRAM

It is the purpose of this Section to outline some of the key features of U. S. undertakings in space. We may undertake space projects for any one of four principal reasons -- scientific; commercial/civilian; military; or national prestige -- although, clearly, these categories overlap. We have accomplished most in the scientific and military area. Our chief lag is, of course, in large scale undertakings chiefly aimed at the enhancement of national prestige. The projects outlined on the attached sheets will be described briefly under each of these headings:

1. Scientific.

The U. S. has accomplished considerably more than the Soviets in the utilization of space technology for the acquisition of new data of scientific importance and value. Our first satellite, EXPLORER I, made possible the initial discovery of the Van Allen radiation belts. Subsequent scientific satellite and probes of the PIONEER and EXPLORER series enabled us to map the Van Allen belts and yielded much additional data. We have learned, and through future experiments will learn a great deal more, about radiation from the sun, from the earth, about magnetic fields in space, the density of cosmic dust, the distribution of meteoric particles, about the size and shape of the earth, and even the size of the solar system by measuring more accurately the distance between space probes and the earth. We are engaged in an extensive program of international cooperation with scientists in numerous other countries and continue to make our scientific findings freely available to the world scientific community. Our program of international cooperation will grow in the years to come.

~~SECRET~~

The national program for the scientific exploration of space has four main approaches. A large number of small payloads are launched each year with sounding rockets. These payloads follow a ballistic trajectory and provide a variety of measurements, of the atmosphere and in space at altitudes to about 4000 miles. Some of these flights will also serve as flight development tests of instruments that are intended for later satellite and deep space probes. This sounding rocket program is expected to continue at a constant rate throughout the next decade.

The second part of the scientific exploration program is based upon measurements with earth satellites, as for example the already discussed PIONEER and EXPLORER series. In the next several years specialized satellites will be launched that will exploit the maximum capabilities of the current U. S. launching vehicles. By 1963, larger vehicles will be available and the scientific satellite program will, for the remainder of the decade, be substantially based on a series of large, general purpose observatory-type spacecraft. Developmental contracts have already been initiated for these observatories: (a) a solar observatory to provide a multiplicity of measurements of interacting sun-earth phenomena; (b) an astronomical observatory to provide an extension of man's observational powers to the whole range of spectra emitted from the universe; and (c) a geophysical observatory to permit simultaneous measurements of the physical quantities in space that are affected by the earth's properties (e.g., atmospheres, ionospheres, magnetic, electrical, and gravitational fields, energetic particles, cosmic rays, etc.). Each of the observatories will contain many individual experiments and will afford an opportunity for experimentation by co-operating scientists throughout the world. Other international experiments of a specialized nature will be conducted using the NASA Scout launch vehicle and NASA sounding rockets.

SECRET

A program of scientific exploration of the moon with instrumented, unmanned spacecraft will constitute the third phase of scientific exploration for the next decade. Within a few months, development flights will be made of a new lunar spacecraft called Ranger. Starting early in 1962 a series of seven of these spacecraft will be launched at the moon. Each will contain photographic equipment capable of recording and transmitting a series of increasingly detailed photographs of the lunar surface as the spacecraft approaches the moon. Shortly before impact, a portion of the spacecraft will be detached and slowed down so that it impacts the moon at a low enough speed to permit instruments to continue to operate on the moon's surface. Several survivable instruments will be included in the over-all program such as seismometers, to measure tremors; penetrometers, to measure the hardness of the surface layer; and, scanning TV to provide close-up details of topography in the vicinity of the impact point.

A second lunar exploration spacecraft, Surveyor, is now under development and a series of ten will be launched beginning in 1963. These will land softly on the moon's surface and will contain a variety of instruments for measuring lunar properties. One of the principal measurements will be a core-sampling through five feet of the moon's crust with on-board analysis of the samples. These results, together with radiation measurements, scanning television pictures, etc., will be transmitted back to the earth.

The last phase of the scientific exploration program is that involving studies of the near-by planets, Venus and Mars, and the intervening space. Spacecraft are now under construction for these missions called Mariner. The initial exploration attempts of the planets, which can only occur at definite intervals when the earth is favorably located, will involve fly-by measurements of the planetary properties. Such an attempt will be made at Venus in the summer of 1962, and again in 1964 when both Venus and Mars are attainable.

Studies are already underway of advanced planetary spacecraft

SECRET

that will be able to go into orbit about Venus and Mars. The planetary satellites will transmit long-term information similar to that now obtained near the earth.

2. Commercial/Civilian.

Vigorous programs are underway to use space for practical, commercial purposes. The outstanding success of the two U. S. meteorological satellites launched to date, TIROS I and TIROS II, has amply demonstrated the utility that space vehicles may be expected to contribute to this area of man's daily concern. The data obtained from these satellites has been used on a semi-operational base by the Weather Bureau, the Air Weather Services of the Department of Defense, and on a more limited basis by a number of participating foreign governments. Sufficient additional TIROS launchings are planned to maintain a continuous source of weather observations using this satellite until a more advanced spacecraft can be employed.

The present TIROS orbit is at a relatively low inclination to the equator, so that observations are only possible in the middle latitude from 50°N to 50°S. The recommended program provides for additional launches to be made on a continuing basis in orbits that will provide weather data over a larger percentage of the earth's surface.

TIROS has several limitations that make it too elementary for a truly operational meteorological system. It is spin-stabilized so that its axis always points in the same direction out into space. As a consequence, the cameras for photographing cloud cover only point at the earth during part of the orbit. In addition, the amount of instrumentation for making measurements of meteorological interest is limited.

These limitations will be eliminated in the next step in the meteorological satellite program which is already underway. A larger, non-spinning spacecraft, NIMBUS, is under development, which will be launched in a polar orbit and which will point at the earth at all times by means of an on-board stabilization system. The first of these

spacecraft will be flown in 1962 and it is possible that sufficient reliability can be achieved to permit a fully operational system by 1963.

Satellite-based telecommunications systems are another application of space technology which will be of great commercial/civilian interest and value in future years. In fact, numerous U. S. commercial enterprises, both equipment manufacturers and common carriers, are greatly interested in this field. Both NASA and the DoD have undertaken projects for the development of communications satellites. Both agencies have studied the subject in depth.

Both agencies have worked closely with the FCC, the OCDM, the White House staff, and other agencies and officials concerned with this subject. Despite the interest evidenced by commercial companies, the technology is not in hand. It is likely that the major preliminary steps leading to the development of commercially feasible communication satellite technology, including the acquisition of environmental data needed to design the satellite vehicle for successful operation, must be done by the Government. Otherwise we must wait until the state-of-the-art generally has developed and deployed on a commercially economical basis.

Accordingly, we have recommended that the NASA Relay project already underway be expanded in FY 62. Within the next few years this country, which is already the world's leader in communications of all kinds, will be able to deploy a worldwide satellite-based communication capability which will enable our friends everywhere to talk with us and among themselves.

The TRANSIT project will permit ships at sea to use radio signals to determine their position within a fraction of a mile. Again, reliability and cost stand in the way of operational deployment. Within a few years this capability, too, will be operationally feasible and ships at sea, whether military or commercial, will be able to utilize the TRANSIT system to determine their position in any kind of weather, night or day.

SECRET

Apart from the important technological byproducts of all phases of our space undertakings, these are the principal operational capabilities which today appear promising for commercial/civilian application during the first half of this decade. Our space effort must include rapid deployment and exploitation of satellites for these purposes.

Space technology, however, is in its infancy. The U. S. conspicuously lacks the capability to place large payloads into orbit. The military applications of such capabilities are not clearly discernable but

SECRET

that does not mean that they will not exist. For this reason, the Department of Defense will undertake the development of large scale segmented solid propellant booster motors. This development will be conducted to meet the requirements of the NASA man-in-space program. The possession of this capacity is considered an important item of insurance for the future from a military point of view.

4. Prestige

A number of U. S. programs, although they are primarily being undertaken for reasons such as scientific knowledge or commercial/civilian values, will have a major worldwide prestige effect. The Ranger and Surveyor projects are designed to give us information about the space near the moon and of the nature and character of the moon surface and interior. Although the Russians have impacted the moon with an object they have not as yet landed instruments on the moon or taken close up pictures of the moon. Our first flights with the Ranger spacecraft are scheduled for calendar 1961. Should we succeed before the Russians substantial prestige value will be ours. The information from the Ranger and Surveyor programs is also an essential prerequisite to manned lunar flights.

Of all the programs planned, perhaps the greatest unsurpassed prestige will accrue to the nation which first sends man to the moon and returns him to earth. The technological advance that will result from the manned lunar landing effort may ultimately overshadow the prestige that will be gained.

The exploration of space will not be complete until man directly participates as an explorer. The national program for manned flight is directed at the earliest accomplishment of this role for man. Project Mercury has already achieved the first milestone of limited, short-duration ballistic flight. By the end of this calendar year it is expected that a four-and-one-half hour flight three times around the earth will have been accomplished.

SECRET

The next major milestone of unsurpassed achievement is the landing of man on the moon. This program, to be known as Project Apollo, has been considered as an active part of NASA planning for the past several years. While the manned lunar landing is an ultimate objective, significant intermediate steps must be undertaken. Information must be acquired in the nature and effect of space radiation on man, man's existence in the hard vacuum of space, and the effects of extended periods of weightlessness on man's performance. Much of the scientific program is geared to the acquisition of this data. Laboratory research has been underway in the nature of and the solution for the intense heating problem during re-entry. The lunar characteristics as a landing site must be understood. The lunar scientific experiments are designed to acquire this information. Finally, adequate launch vehicles must be developed, not only for the final landing mission, but for the intermediate developmental steps. These will include manned orbital flight around the earth for extended periods, and manned circumlunar flight.

The extended orbital flight will be accomplished for periods up to 27 hours (18-orbits) using modified Mercury capsules. The necessary longer duration (2 weeks) flights will utilize the Saturn C-1 launch vehicle and a version of the final lunar landing capsule. Prior to these flights, it will be necessary to fly primates for extended periods. They will be done with modified Mercury capsules.

The manned circum^mlunar flight must have a greater launch vehicle capability than the earth orbital flights. This may be provided by the Saturn C-2 vehicle. The final lunar landing will require a launching vehicle from earth that is about six times as big as Saturn. NASA has the 1.5 million lb. thrust F-1 engine under development. This engine is expected to power a Nova vehicle. At the same time a parallel development leading towards large launch vehicles based on solid propellants will be undertaken. As one or the other approach proves superior that approach will be selected for the NOVA vehicle. The DoD will carry out the solid rocket development with full responsiveness to NASA requirements.

SPACE ACTIVITIES OF THE UNITED STATES GOVERNMENT
New Obligational Authority/Program Basis - in millions

Historical Summary and Proposed 1962 Add-ons

	<u>NASA</u> ^{1/}	<u>Defense</u> ^{2/}	<u>AEC</u> ^{3/}	<u>NSF</u> ^{4/}	<u>WB</u> ^{5/}	<u>Total</u>
1955	\$56.9	\$ 3.0	-	-	-	\$ 59.9
1956	72.7	30.3	\$ 7.0	\$7.3	-	117.3
1957	78.2	71.0	21.3	8.4	-	178.9
1958	117.3	205.6	21.3	3.3	-	347.5
1959	338.9	489.5	34.3	-	-	862.7
1960	523.6	560.9	43.3	.1	-	1,127.9
1961	964.0	751.7	62.7	.6	-	1,779.0
1962 Budget	1,109.6	846.9	55.1	1.6	\$2.2	2,015.4
Budget amendments .	<u>125.7</u>	<u>159.0</u>	<u>23.5</u>	<u>-</u>	<u>-</u>	<u>308.2</u>
Total present 1962 Budget ..	1,235.3	1,005.9	78.6	1.6	2.2	2,323.6
Proposed 62 Add-ons	<u>549.0</u>	<u>77.0</u>	<u>7.0</u> ^{6/}	<u>-</u>	<u>53.0</u> ^{7/}	<u>686.0</u>
Total 1962 Proposals	\$1,784.3	\$1,082.9	\$85.6	\$1.6	\$55.2	\$3,009.6

1/ National Aeronautics and Space Administration amounts are totals for all activities of NASA and include totals for NACA prior to establishment of NASA.

2/ Department of Defense amounts are based on identifiable Defense funding for space and space-related effort and do not include substantial amounts for (1) construction and operation of the national missile ranges with regard to space programs, (2) the cost of developing missiles such as Thor and Atlas which are also used in space programs, or (3) supporting research and development (such as bio-medical research) which is more or less mutually applicable to programs other than "space".

3/ Atomic Energy Commission amounts are those identifiable with ROVER nuclear rocket and SNAP atomic power source projects.

4/ National Science Foundation amounts are those identifiable with VANGUARD and with the NSF space telescope project.

5/ Weather Bureau amounts are those identifiable with the meteorological satellite program.

6/ AEC add-on for ROVER corresponding to the \$23 million included in proposed NASA add-ons.

7/ Add-on for initiation of operational NIMBUS system under consideration by Department of Commerce.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
FY 1962 BUDGET AND PROPOSED ADD-ONS

	(In millions)				
	Original 1962 Budget	1962 Budget Amendment	Present 1962 Budget	Proposed 1962 Add-ons	Total 1962 Proposed
I.. Major Vehicle & Propulsion					
Dev. Projects					
A. Centaur	\$ 49.8	\$ 25.6 ^{1/}	\$ 75.4	-	\$ 75.4
B. Saturn	212.8	73.0 ^{1/}	285.8	-	285.8
C. Rover (NASA only)	13.5	4.0	17.5	\$ 23.0 ^{4/}	40.5
D. F-1 Engine and Nova Vehicle	33.4	9.3	42.7	112.5	155.2
E. Large Solid Boosters ...	2.0	-	2.0	-	2.0
F. Other Projects	6.6	-	6.6	-	6.6
II. Major Space Flight Programs					
A. Mercury	74.2	-	74.2	-	74.2
B. Other Manned Space Flight	29.5	-	29.5	210.5	240.0
C. Meteorology	28.2	-	28.2	22.0	50.2
D. Communications	34.6	10.0	44.6	50.0	94.6
E. Scientific Satellites & Sounding Rockets	72.2	-	72.2	15.0	87.2
F. Unmanned Lunar Exploration	71.7	-	71.7	63.0	134.7
G. Unmanned Planetary Exploration	32.2	-	32.2	-	32.2
III. Supporting Development and Operations					
A. NASA Development Centers & Launch Operations ..	164.0	3.0 ^{2/}	167.0	28.0 ^{5/}	195.0
B. Tracking and Data Acquisition	60.2	-1.0	59.2	-	59.2
C. Other Supporting Development	57.7	-	57.7	25.0	82.7
IV. Aeronautical and Space Research					
.....	142.7	1.3 ^{3/}	144.0	-	144.0
V. Program Direction					
.....	24.3	.5	24.8	-	24.8
Total NASA	\$1,109.6	\$125.7	\$1,235.3	\$549.0	\$1,784.3

^{1/} Total of \$78 million identified with Saturn C-2 includes \$5 million of supporting costs on line III-A.

^{2/} Consists of \$5 million for support identified with Saturn C-2 less \$2 million adjustment in other supporting costs.

^{3/} Consists of \$2 million identified with the supersonic transport less \$.7 million adjustment on a vivarium construction project.

^{4/} Would require corresponding add-on of \$7 million for AEC.

^{5/} Part of this amount may be applied to IV and V.